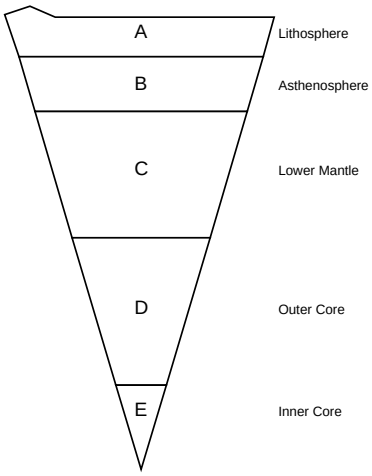
Plate Tectonics & Volcanoes -- Answer key

Name: _____



1. Wegener's theory: Continental Drift
2. His two clues: Matching mountains; tropical plant fossils in the Arctic
3. Hess's tool: Sonar
4. At a mid-ocean ridge: Plates split apart
5. Asthenos means: weak

6. Describe convergent, transform, and divergent movement.
   Convergent: together. Transform: slide past. Divergent: apart.

7. What happens during subduction?
   A denser ocean plate slides beneath another plate.

8. Which layer is hottest? How do magma and lava differ?
   Inner Core. Magma: below the surface; lava: on the surface.

9. Name two Ring of Fire boundaries (both plates for each).
   Nazca / South American; Pacific / Australian.

10. What is a dormant volcano?
    Quiet for a long time but may erupt again.

11. Compare composite and shield volcanoes: eruptions, magma, shape, location.
    Composite: short explosive, sticky/high viscosity, steep, subduction. Shield: long quiet, thin/low viscosity, gentle slopes, hot spots.

12. Name three shared volcano traits from the sheet.
    Erupt; produce lava and cinders; produce great soil.

13. Why is plate tectonics like a recycler?
    New crust forms at divergent boundaries; old crust sinks at subduction zones and its material is recycled into the mantle.

Answer key uses worksheet study targets. See Learn for science notes and explanations.